

Foundation (Jalapeno)

Q1. Solve the equation $|x| = 5$

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve the equation $|2x| = 8$

- (A) $x = 2$
- (B) $x = -2$
- (C) $x = 2$ or $x = -2$
- (D) $x = 4$
- (E) $x = -4$

Q3. Solve the equation $|x + 2| = 3$

- (A) $x = 1$
- (B) $x = -1$
- (C) $x = 1$ or $x = -5$
- (D) $x = -3$
- (E) $x = 5$

Q4. Solve the equation $|x - 3| = 2$

- (A) $x = 1$
- (B) $x = 5$
- (C) $x = 1$ or $x = 5$
- (D) $x = 3$
- (E) $x = 0$

Q5. Solve the equation $|3x| = 12$

- (A) $x = 2$
- (B) $x = -2$
- (C) $x = 2$ or $x = -2$
- (D) $x = 4$
- (E) $x = -4$

Q6. Solve the equation $|x + 1| = 4$

- (A) $x = 3$
- (B) $x = -3$
- (C) $x = 3$ or $x = -5$
- (D) $x = -1$
- (E) $x = 5$

Q7. Solve the equation $|2x - 1| = 7$

- (A) $x = 2$
- (B) $x = -2$
- (C) $x = 4$ or $x = -3$
- (D) $x = 3$
- (E) $x = -4$

Q8. Solve the equation $|x + 4| = 1$

- (A) $x = -3$
- (B) $x = -5$
- (C) $x = -3$ or $x = -5$
- (D) $x = -1$
- (E) $x = 3$

Open Ended Questions (FRQ)

Q9. Solve the equation $|x^2 - 4| = 8$ and check for extraneous solutions

Q10. Solve the equation $|3x + 2| = |2x - 5|$ and check for extraneous solutions

Chef's Tips

- Check for extraneous solutions
- Use the definition of absolute value to solve equations
- Consider all possible cases when solving absolute value equations